n=47 random00: the degree-four closure equals the accepted quotient kernel

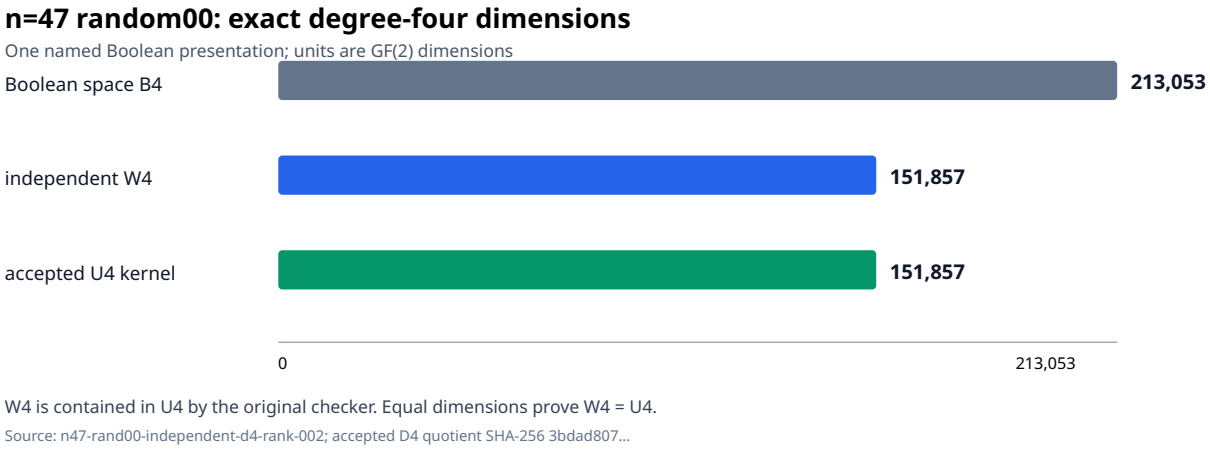
Exact finite-instance evidence record

2 November 2026

The independent implementation B replay returned degree-four Boolean closure rank 151,857 in B_4 of dimension 213,053, with $1 \notin W_4(F)$. The original independent D4 checker accepted a surjective quotient with 61,196 coordinates and proved $W_4(F) \subseteq U_4$. Therefore

$$\dim U_4 = 213{,}053 - 61{,}196 = 151{,}857 = \dim W_4(F), \qquad W_4(F) = U_4.$$

The conclusion applies to the single frozen n=47 random00 Boolean presentation. The replay reported 12,535 seconds under a 16 GiB child address-space cap; no comparative timing claim is made.



The frozen plan pins the field, curve, target, subspace, accepted seed hash, checker transcript and both source hashes. The archived output has status `rank_match`, return code 0, no timeout, and empty stderr. `audit.py` verifies every archived output digest and, with the local seed, the 1.63 GB seed hash and header. The audit does not rerun the exact closure.

This closes the D4 seed-equality obligation for this input. The earlier D5 producer reported a killed constant but retained no independently checkable terminal relation witness. Hence this report establishes $U_4 = W_4$ and leaves the D5 membership/upper-bound question open. A checked D5 primal derivation or independent full relation replay is the next proof step.

Evidence: the frozen plan, archived result and checker transcript, evidence manifest, and audit script in this report directory. The D4 seed SHA-256 begins `3bdad807113e43ec`; the complete digest is in the plan and manifest.